MIL-P-51272
22 August 1968
(See section 6)

MILITARY SPECIFICATION

PALLADIUM CHLORIDE, ANHYDROUS, REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one reagent grade of anhydrous palladium chloride ($PdCl_2$).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Appearance. Palladium chloride shall be a dark brown powder, when examined as specified in 4.2.4.1.

3.2 Assay (as palladium). Palladium chloride shall contain no less than 59.4 percent by weight of palladium (Pd) when tested as specified in 4.2.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the palladium chloride produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation of delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. Unless otherwise specified (see 6.2), a representative specimen of approximately 5 grams shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit or shipping container, as applicable. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective.

(a) Contents per container not as specified

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(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 5 grams. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. The tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine 2 grams (g) of the specimen for color and form.

4.2.4.2 Assay (as Pd). Weigh to the nearest milligram (mg) approximately 0.25 g of the specimen. Dissolve in 10 milliliters (ml) of warm 10-percent hydrochloric acid. Dilute with water to 100 ml. Add 100 ml of 1-percent alcoholic dimethylglyoxime and heat on a steam bath for 1 hour. Filter while warm through a tared Gooch crucible, using suction. Wash the precipitate, first with 1-percent hydrochloric acid, then with hot water. Dry at 105° ± 5° C to a constant weight. Cool in a desiccator. Weigh to the nearest 0.1 mg. Calculate the percent by weight of palladium as follows:

$$\text{Assay, as percent palladium} = \frac{31.67(A-B)}{W}$$

where: A = Weight of the crucible and precipitate in grams,
B = Weight of the crucible in grams, and
W = Weight of the specimen in grams.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Ten ± 0.05 g of palladium chloride shall be packaged in a glass bottle conforming to type I, class 1 of PPP-C-301. Inclosure b shall be used.

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5.1.2 Level C. Unless otherwise specified in the contract or order (see 6.2), a 10 g ± 0.05 g quantity of palladium chloride shall be packaged to provide adequate protection against contamination, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Twelve bottles of palladium chloride, packaged as specified in 5.1.1, shall be packed in a wood box conforming to class 2, style 4-1/2 for type 2 load of PPP-B-621. All inside faces of the box shall be lined with fiberboard conforming to grade W6c of PPP-F-320. The box shall be closed and strapped with galvanized strapping as specified in the appendix to the box specification.

5.2.2 Level B. Palladium chloride shall be packed as specified in 5.2.1, except that a fiberboard box conforming to style RSC, grade W5c of PPP-B-636 shall be used. The box shall be closed in accordance with the requirements for taping style RSC boxes to be shipped as exterior containers, as given in PPP-B-636. The closed box shall be reinforced by tape banding, as specified in the appendix to the box specification.

5.2.3 Level C. Palladium chloride, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers acceptable to the mode of transportation.

5.3 Marking. In addition to the special marking for unit containers specified in 5.3.1, and any special marking specified by the contract or order (see 6.2), unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following information:

(a) Net weight of contents.
(b) Empirical formula.
(c) Formula weight.
(d) Manufacturer's lot number.
(e) Assay.

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6. NOTES

6.1 Intended use. Palladium chloride is intended for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing.
(c) Quantity required for level C packaging, if other than specified.
(d) Special marking other than 5.3.1, if required.
(e) Whether smaller individual specimens for sampling are permitted.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded of "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Supersession data. This specification includes the requirements of Edgewood Arsenal Purchase Description 4-145, dated 3 December 1952.

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0451

☆ U. S. GOVERNMENT PRINTING OFFICE: 1968-341-511/A-1357

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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Commanding Officer
Edgewood Arsenal
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